Digital Signal Processing
ECG FIR Filtering Assignment

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1 Plotting the ECG Signal

The experiment began by taking an ECG recording and plotting it in python using matplotlib. The electrodes were attached just above the hip and on the front of the shoulder.

1.1 Structure of PQRST Complex

It is important that we preserve the frequencies used to construct the PQRST complex, otherwise we would be destroying important data from the ECG.

1.2 Sources of Noise in the ECG

The main sources of noise in an ECG are 50Hz mains interference, Breathing, and random noise caused by movement. There is also a DC offset which shifts the entire signal by a certain amount, caused by potential difference formed between the electrodes and the skin.
The human heart rate can go as low as 0.5Hz at complete rest. Considering this, the filter should remove frequencies below 0.5Hz (This includes DC shift and Breathing) and above 45Hz (Mains noise and other high frequency interference). Although this leaves in random noise cause by movement, the core frequencies of the PQRST complex must be retained for the signal to remain accurate for monitoring purposes.

2 Generating the FIR Coefficients

2.1 Stop-Band Filter

A stop-band filter was implemented in this code by utilizing an all-pass filter, a band-pass filter, and the principle of superposition. This was done instead of using a pure band-stop filter in order to keep the code simple and elegant, as it is easier to remove frequencies from an all-pass filter than it is to combine multiple separate lo-pass and hi-pass filters together.
First, the number of coefficients needed for the windowing function to work had to be calculated. For a Kaiser Window, this changes based on the desired stopband attenuation and transition width. The formula is displayed in the next subsection in Equation (3). Then, using the number of taps, an ideal all-pass filter was defined by creating an array of zeros, with an index stretching from indices of -m to m (where m = num_taps / 2), with the central value at index m set to 1. This represents the ideal impulse response. Finally the array n is initialized, which represents discrete time indices from -m to m. Evalutating functions over this symmetric time vector is what gives the filter linear phase and constant group delay.
This all-pass filter is then modified in a for loop iterating over the array of stop bands. The stop bands live in the frequency domain, so an inverse discrete time fourier transform is applied to convert them into the time-domain before they are subtracted from the impulse response:

$$h_{\text{band}}[n] = 2 \cdot f_2 \cdot \text{sinc}(2 \cdot f_2 \cdot n) - 2 \cdot f_1 \cdot \text{sinc}(2 \cdot f_1 \cdot n) \tag{1}$$

(Note: Doing the IDFT manually is less efficient than using np.IFFT() and was done purely for analytical purposes)

After this for-loop is complete, h will be a raw band-stop filter (sharp edges in the frequency domain, causing ringing due to the Gibbs Phenomenon). The Gibbs phenonenom can be seen in the impulse response of the filter as the side lobes aren't able to settle to zero before they get truncated:

To fix this, it must be multiplied with a windowing function.

2.2 The Kaiser Window

The Kaiser window was chosen for its flexibility and simplicity. It gives greater flexibility than other windowing functions, since the user has control over both the size of the transition window and the stopband attenuation. This allows you to find a precise balance between a narrow central peak (sharper cutoff) and flatter sidelobes (better noise rejection) in the impulse response. With specific stopband attenuations, it is also possible for the Kaiser window to mimic other windowing functions. Furthermore, the Kaiser filter has straightforward formulas to calculate beta (the stopband attenuation parameter) and num_taps (the number of coefficients needed for the filter), allowing it to be practically calculated almost entirely in code without external libraries. The formulae used in the code are as follows:

beta parameter:

$$\beta = \begin{cases} 0.1102(A - 8.7) & \text{if } A > 50 \\ 0.5842(A-21)^{0.4} + 0.07886(A-21) & \text{if } 21 \le A \le 50 \\ 0 & \text{otherwise} \end{cases} \tag{2}$$

number of taps:

$$N = \left\lfloor \frac{A - 8}{2.285 \cdot 2\pi \left(\frac{\Delta f}{f_s}\right)} \right\rfloor \tag{3}$$

kaiser window:

$$w[n] = \frac{I_0\left(\beta\sqrt{1 - \left(\frac{n}{m}\right)^2}\right)}{I_0(\beta)} \tag{4}$$

The FIR filter is then multiplied by the Kaiser window, to create the final array of coefficients, producing this impulse response in the time domain:

As you can see, this impulse response has a large, narrow central peak and sidelobes that gradually decay to zero as they approach tap 0 and tap 383. This shows that our windowing function has successfully removed ringing due to the Gibbs Phenomenon. These coefficients can now be passed into the fir_filter class to simulate realtime filtering.

3 Simulation of Real-time Filtering

In order to simulate realtime filtering of the data, a python class was created to represent the FIR class. This class takes the taps (calculated from the filter design function) as it's sole argument. Additionally, the class has 3 data members; coefficients is an array of the filter's taps; num_taps is the length of coefficients; And delay_line is a double ended queue with a maximum length of num_taps. A double ended queue was used as the data structure for the delay line because, when given a maximum length, it acts similarly to a ring buffer where adding a new value to a full queue causes one value to fall off the other end. This is an efficient implementation, because it does not require manual shifting of the delay line.
The function member used for simulating realtime processing is simply dofilter. This takes a scalar input of the newest sample in the data, appends it to the delay line and then loops through the delay line to calculate a cumulative sum of delay_line[i] * coefficients[i]. It returns the result of this summation. To simulate realtime input to the filter in python, a for loop iterates through the ecg data and passes it sample by sample to the FIR filter for processing.

3.1 Filter Design

$$MathBlock \tag{5}$$

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4 Adaptive LMS Filter

The doFilterAdaptive function member extends the FIR filter class for adaptive filtering. The arguments to this function are signal, noise, and learningRate. The signal argument is the signal that we are trying to remove from our ecg. The noise argument is our noisy ecg signal, and learningRate is the learning rate/step size of our adaptive algorithm. The function initially calls the doFilter function on the signal for the forward pass before updating the weights of the filter using a least mean squares method:

Our signal input is a 50Hz sin wave plus a DC shift of 0.25. The value of the DC shift was found experimentally with the goal of removing the DC shift from

$$error = noise - filtered_signal \tag{6}$$

$$coefficients[i] = coefficients[i] + learningRate \cdot error \cdot delay_line[i] \tag{7}$$

the signal without degrading the quality of the PQRST wave. The learning rate was also found experimentally such that the filter coefficients converge quickly without exploding.

4.1 Comparison with FIR Filter

5 Heartbeat Detection

5.1 Subsection

6 Appendix

6.1 ECG Plotting Function

```
import numpy as np
from pathlib import Path
import matplotlib.pyplot as plt

IMAGES = Path("images")
IMAGES.mkdir(parents = True, exist_ok = True)

file_name = "James_ecg_1.tsv"
tsv = np.loadtxt(file_name, delimiter='\t')
ecg_signal = tsv[:,8].copy()


sample_rate = 250
num_samples = len(ecg_signal)
measurement_time = num_samples / sample_rate
t = np.linspace(0, measurement_time, num_samples)

plt.figure()
plt.title("time_domain_ecg")
plt.plot(t,ecg_signal)
plt.savefig(IMAGES / f"time_domain_{file_name}.svg")
plt.show()
```

6.2 Windowed Band-Stop filter Implementation

```
import numpy as np

# sampling_rate, frequency ranges to block, desired stop-band attenuation, width of transition window
def getTaps(sampling_rate, stop_bands, A, transition_width):

    # 1. Calculate the stopband attenuation and filter order for the kaiser window
    if A > 50:
        beta = 0.1102 * (A - 8.7)
    elif 21 <= A <= 50:
        beta = 0.5842 * (A - 21) ** 0.4 + 0.07886 * (A - 21)
    else:
        beta = 0

    # Formula for num_taps of Kaiser window: (A - 8) / (2.285 * d_omega), where d_omega is the normalized transition width in radians
    num_taps = int((A - 8) / (2.285 * 2 * np.pi * (transition_width / sampling_rate)))

    if num_taps % 2 == 0:
        num_taps += 1 # Ensure the number of taps is odd so the filter order will be even

    # 2. Define the impulse Response of the ideal frequency response from stop_bands
    m = (num_taps - 1) // 2
    h = np.zeros(num_taps)
    h[m] = 1 # Initialize h as an all-pass filter (unit impulse)
    n = np.arange(-m, m + 1)

    for f1, f2 in stop_bands:
        # Normalize Frequencies
        f1 = f1 / sampling_rate
        f2 = f2 / sampling_rate

        # IFFT Formula for bandpass filter: h(n) = 1 / pi * n * [sin(w2 * n) - sin(w1 * n)]
        # Simplified using sinc function: h(n) = [2 * f2 * sinc (2 * f2 * n)] - [2 * f1 * sinc (2 * f1 * n)]
        h_band = 2 * f2 * np.sinc(2 * f2 * n) - (2 * f1 * np.sinc(2 * f1 * n))

        # Principle of superposition - subtract bandpass filter from allpass filter to create a bandstop filter
        # WARNING: Ensure stop-band ranges don't overlap to avoid signal distortion
        h -= h_band

    # 3. Calculate the Kaiser Window function using a modified bessel function of the first kind
    # Kaiser window formula: I0 [beta * sqrt(1 - {(n - alpha) / alpha} ** 2)] / I0 * beta
    kaiser_window = np.i0(beta * np.sqrt(1 - (n / m) ** 2)) / np.i0(np.array([beta]))[0]

    # 4. Window the stop-band filter with our kaiser window
    taps = h * kaiser_window

    return taps, num_taps
```

6.3 FIR Filter Implementation

```
import collections
```

```
class FIRfilter:
    def __init__(self, _coefficients):
        # Store the coefficients in this filter object (denoted by "self.")
        self.coefficients = _coefficients
        # Store the number of taps (== length of coefficients array)
        self.num_taps = len(self.coefficients)
        # Initialize a double-ended queue with a length of num_taps
        self.delay_line = collections.deque(maxlen=self.num_taps)

        return

    def dofilter(self, v):
        # Append incoming samples to the first index of the double-ended queue
        self.delay_line.appendleft(v)
        # initialize an accumulator for this window
        result = 0

        # Perform a convolution sum - multiply each sample in the buffer by its corresponding coefficient,
        # and add the result to the accumulator
        for i in range(len(self.delay_line)):
            result += self.delay_line[i] * self.coefficients[i]

        return result

    def dofilteradaptive(self, signal, noise, learningRate):
            """
            Adaptive filtering using the LMS algorithm.
            - signal: The input signal, to be filtered out
            - desired: The desired signal to be used for adaptation AKA ecg
            - mu: The step size for adaptation
            - result: The filtered signal
            """

            # perform a convolution sum to modify the amplitude and phase of the pure input signal
            filtered_signal = self.dofilter(signal)

            # calculate the error (the clean signal)
            error = noise - filtered_signal
            for i in range(len(self.delay_line)):
                # Update coefficients using LMS algorithm
                self.coefficients[i] += learningRate * error * self.delay_line[i]

            # Return the error signal (the difference between the desired signal and the filtered output)
            return error
```

6.4 Realtime Filtering Simulation Implementation

```
import fir_design as fir_design
import fir_filter as fir_filter
import numpy as np
import matplotlib.pyplot as plt
from pathlib import Path

tsv = np.loadtxt("James_ecg_1.tsv", delimiter='\t')
ecg_signal = tsv[:,8].copy()

IMAGES = Path("images")
IMAGES.mkdir(parents = True, exist_ok = True)

# Performing FIR Filtering

num_samples = len(ecg_signal) # Number of Samples in ECG signal
sample_rate = 250 # Sampling rate of ECG signal (Hz)
width = 1 # desired width of the transition window
stop_Hz = [(0, 3), (40, sample_rate // 2)] # The cutoff frequency for the filter
A = 30 # stopband attenuation

# get the array of taps and number of taps (taps == coefficients)
taps, num_taps = fir_design.getTaps(sample_rate, stop_Hz, A, width)

# use FIRfilter to filter x with the FIR filter
filter = fir_filter.FIRfilter(taps)

# Simulate realtime filtering of the ECG
filtered_ecg_signal = np.zeros(num_samples)
for i in range(num_samples):
    filtered_ecg_signal[i] = filter.dofilter(ecg_signal[i])
```

```
measurement_time = num_samples / sample_rate
t = np.linspace(0 , measurement_time , num_samples)

plt.figure()
plt.title("ecg_filtered_time_domain")
plt.plot(t, filtered_ecg_signal)
plt.savefig(IMAGES / "ecg_filtered_time_domain.svg")
```

6.5 Adaptive Filter Implementation

```
import fir_design as fir_design
import fir_filter as fir_filter
import numpy as np
import matplotlib.pyplot as plt
from pathlib import Path

IMAGES = Path("images")
IMAGES.mkdir(parents = True, exist_ok = True)

tsv = np.loadtxt("James_ecg_1.tsv", delimiter='\t')
ecg_signal = tsv[:,8].copy()

# This data is not used for the adaptive filtering, but allows us to call getTaps to get the number of taps for the adaptive filter.
# The adaptive filter will be initialized with all coefficients set to zero.
num_samples = len(ecg_signal) # Number of Samples in ECG signal
sample_rate = 250 # Sampling rate of ECG signal (Hz)
width = 1 # desired width of the transition window
nyquist_freq = sample_rate // 2
stop_Hz = [(0, 3), (45, nyquist_freq)] # The cutoff frequency for the filter
A = 30 # stopband attenuation

# get the number of taps (taps == coefficients).
taps, num_taps = fir_design.getTaps(sample_rate, stop_Hz, A, width)

#Simulate realtime adaptive filtering of the ECG

# Initialize adaptive filter with num_taps coefficients all set to zero
adaptivefilter = fir_filter.FIRfilter(np.zeros(num_taps))

# Initialize an array to hold the adaptive filtered ECG signal
adaptive_filtered_ecg_signal = np.zeros(num_samples)

# Use 50Hz as the signal for adaptive filtering (to be filtered out)
fiftyHz = np.sin(2 * np.pi * 50 / sample_rate * np.arange(num_samples)) + 0.25

# Use the filtered ECG signal as the noise reference for adaptive filtering
for i in range(num_samples):
    adaptive_filtered_ecg_signal[i] = adaptivefilter.dofilteradaptive(fiftyHz[i], ecg_signal[i], learningRate=0.001)

measurement_time = num_samples / sample_rate
t = np.linspace(0 , measurement_time , num_samples)

plt.figure()
plt.title("ecg_adaptive_filtered_time_domain")
plt.plot(t[0:1000], adaptive_filtered_ecg_signal[0:1000])
plt.savefig(IMAGES / "ecg_adaptive_filtered_time_domain.svg")
plt.show()
```

6.6 Heartbeat detection

```
import fir_design as fd
import fir_filter as ff
import numpy as np
import matplotlib.pyplot as plt
from pathlib import Path

IMAGES = Path("images")
IMAGES.mkdir(parents = True, exist_ok = True)

tsv1 = np.loadtxt("James_ecg_1.tsv", delimiter='\t')
ecg_signal = tsv1[:,8].copy()

# Performing FIR Pre-Filtering to remove DC offset, baseline wander, and 50Hz mains interference

num_samples = len(ecg_signal) # Number of Samples in ECG signal
sample_rate = 250 # Sampling rate of ECG signal (Hz)
width = 1 # desired width of the transition window
stop_Hz = [(0, 3), (40, sample_rate // 2)] # The cutoff frequency for the filter
A = 30 # stopband attenuation
```

```
# get the array of taps and number of taps (taps == coefficients)
taps, num_taps = fd.getTaps(sample_rate, stop_Hz, A, width)

# use FIRfilter to filter x with the FIR filter
prefilter = ff.FIRfilter(taps)

# Simulate realtime filtering of the ECG
filtered_ecg_signal = np.zeros(num_samples)
for i in range(num_samples):
    filtered_ecg_signal[i] = prefilter.dofilter(ecg_signal[i])

tsv2 = np.loadtxt("Ewan_ecg_1.tsv", delimiter='\t')
ecg_template = tsv2[:,8].copy()

# use new prefilter to filter the ECG template signal. So delay line is empty
prefilter2 = ff.FIRfilter(taps)

# Simulate realtime filtering of the ECG template
filtered_ecg_template = np.zeros(len(ecg_template))
for i in range(len(ecg_template)):
    filtered_ecg_template[i] = prefilter2.dofilter(ecg_template[i])

#extract one PQRST complex from the filtered ECG template for matching
filtered_ecg_template = filtered_ecg_template[500:700]

# Reverse the filtered ECG template for cross-correlation
filtered_ecg_template = filtered_ecg_template[::-1]

matched_filter = ff.FIRfilter(filtered_ecg_template)

matched_filtered_ecg_signal = np.zeros(num_samples)

num_samples_template = len(matched_filtered_ecg_signal) # Number of Samples in ECG signal
measurement_time = num_samples_template / sample_rate
t2 = np.linspace(0 , measurement_time , num_samples_template)

#Realtime matched filtering of the filtered ECG signal with the filtered ECG template
last_peak_index = 0
current_peak_index = 0
heartrate = []
cooldown_period = 68  # 68/250Hz - limits heartrate to 220bpm
threshold = 3*10**-11

cooldown = 0
for i in range(num_samples):
    matched_filtered_ecg_signal[i] = matched_filter.dofilter(filtered_ecg_signal[i])

    # Square the matched filtered signal to enhance the peaks for easier detection of R-peaks
    matched_filtered_ecg_signal[i] = matched_filtered_ecg_signal[i] * matched_filtered_ecg_signal[i]

    if cooldown > 0:
        cooldown -= 1

    elif matched_filtered_ecg_signal[i] > threshold:
        last_peak_index = current_peak_index
        current_peak_index = i

        if last_peak_index > 0: # Ensure there are at least two beats before calculating heartrate
            heartrate.append(60 * sample_rate / (current_peak_index - last_peak_index))

        cooldown = cooldown_period


plt.figure()
plt.title("ecg_matched_filtered_time_domain")
plt.plot(t2, matched_filtered_ecg_signal)
plt.savefig(IMAGES / "ecg_matched_filtered_time_domain.svg")

plt.figure()
plt.title(IMAGES / "heart_rate_time_domain")
plt.plot(heartrate)
plt.savefig(IMAGES / "heart_rate_time_domain.svg")

plt.show()
```